

HANAMANTHAGOUDA POLICEPATIL

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EDUCATION

Indian Institute of Technology Gandhinagar (IITGN)
Dual B.Tech. in Computer Science and Materials Science & Engineering

Jul '23 – May '28
CPI: 8.0 / 10.0

WORK EXPERIENCE

Research Intern | Indian Institute of Science (IISc), Bengaluru

May '25 – Jul '25 (Onsite)

Advisor: Prof. Abhishek Kumar Singh | Materials Research Centre (MRC)

Bengaluru, India

- **Multimodal Architecture:** Designed domain-adaptive foundation model pipelines (CAST, CLIP, DAPT) unifying unstructured scientific literature with structural crystal lattice data.
- **Instruction Fine-Tuning:** Fine-tuned LLaMA-3 8B using QLoRA and Unsloth on specialized instruction datasets; optimized context length and hyperparameters, increasing classification accuracy from a 52.9% baseline to **90.2%**. [\[GitHub\]](#)
- **Domain-Adaptive Pre-Training (DAPT):** Executed target-domain CPT on a multi-million-token corpus using Unsloth (test perplexity **9.48**); built automated benchmarks and integrated checkpoints onto Hugging Face. [\[GitHub\]](#)

PUBLICATIONS

Real-Time Win Probability Estimation in Twenty20 Cricket: A State-Space Approach

2025

Co-authors: Prof. Prateek Sharma (USC) & Prof. Amit Arora (IITGN) | Submitted to *Journal of Quantitative Analysis in Sports* | [\[SSRN\]](#)

- Formulated a state-space framework with recursive **Kalman filtering** to model latent in-game dynamics (team pressure, momentum shifts), bridging sequential probabilistic forecasting with interpretable match trajectories.
- Benchmarked against tree-based ensembles (XGBoost, Random Forests) and logistic regression across **1,161 historical matches**; demonstrated superior predictive capability during rapid in-game turning points.

PROJECTS

eBPF Real-Time Network Bandwidth Tracker

Sep '26

Linux Kernel Probing | eBPF (BCC), C, Python, SQLite WAL, FastAPI CS331: Computer Networks | [\[GitHub\]](#)

- **Kernel-Level Telemetry:** Hooked Linux socket probes (kprobe/tcp_sendmsg, tcp_cleanup_rbuf, udp_sendmsg) using BCC to capture per-flow socket throughput in kernel space with zero context-switch overhead.
- **Process Attribution & Storage:** Streamed flow metrics through perf ring buffers to a Python daemon; attributed traffic to PIDs, process names, and remote IPs via a TTL cache, persisting time-series metrics in SQLite WAL.
- **Precision & Interfaces:** Verified **100.00% byte accuracy** under saturating network loads (> 800 Mbps) with negligible (< 0.01%) CPU overhead; built an interactive terminal TUI dashboard and a FastAPI web visualizer.

Scientific Machine Learning for Physical Systems (SciML)

2024-2

Deep Learning for Physical Systems | PyTorch, SciPy

IIT Gandhinagar

- **Fourier Neural Operators (FNOs):** Implemented FNO in PyTorch to simulate 2D Navier-Stokes turbulence. Integrated targeted noise injection during autoregressive rollouts, stabilizing long horizons to **1.04%** validation error. [\[GitHub\]](#)
- **Physics-Informed Neural Networks (PINNs):** Formulated PINNs for 1D Burgers' equation forward field solving and inverse latent parameter estimation (λ_1, λ_2) from noisy measurements with under **0.22%** error. [\[GitHub\]](#)

SKILLS

Programming Languages: Python, C++, C, SQL, Bash

Machine Learning & Libraries: PyTorch, Hugging Face, Scikit-Learn, Unsloth, NumPy, Pandas, SciPy, Matplotlib

Developer Tools & Systems: Git, Linux/Unix, Docker, eBPF (BCC), SQLite, FastAPI

LEADERSHIP

Class Representative (MSE '27) | IIT Gandhinagar

2024-25

Primary academic liaison for 40+ undergraduate students; resolved curriculum conflicts and coordinated with faculty.